



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

EXHUAUSTIVE ANALYSIS OF INDIAN AGRICULTURAL SECTOR USING POWER BI

The domain of the Project:

POWER BI & SQL

Team Mentors (and their designation):

SRAVAN NEMANA

Team Members:

Mr. Ankamma Bellamkonda

Period of the project

Nov 2025 to Dec 2025



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Declaration

The project titled “Exhaustive analysis of Indian Agricultural sector using Power BI” has been mentored by Sravan Nemana, organised by SURE Trust, from April 2023 to August 2023, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Mr. Ankamma Bellamkonda

Mentor's Name

Sravan Nemana

Designation—Company Name

Co-founder – Stealth AI startup

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

- This project, titled "Exhaustive Analysis of Indian Agriculture Sector Using Power BI," provides a comprehensive data-driven exploration of crop production trends in India.
- Given that agriculture contributes over 18% to India's GDP, this analysis offers real-world insights into a critical economic sector.
- The primary objective was to uncover key trends, productivity insights, and identify high-performing states and crops using a dataset comprising 73,827 records.
- The project utilized Power BI to develop interactive dashboards that visualize the relationships between production and cultivated area.
- Key findings highlight production trends across different regions, crops, and seasons, identifying top-performing crops and high-yield states.
- These insights are designed to support policymakers and researchers in making data-driven decisions.
- By leveraging Microsoft Power BI, the team transformed raw data into interactive dashboards. These visualizations highlight the relationship between the area cultivated and the total production, calculating key performance indicators (KPIs) such as Average Yield and Total Cultivated Area.
- The outcome of this project provides actionable insights that can support policymakers, researchers, and stakeholders in making informed, data-driven decisions to enhance agricultural productivity in India.
- The project utilized a robust dataset comprising covering essential attributes such as State names, District names, Crop Years, Seasons, Crops, Cultivated Area, and Production figures.
- The primary objective was to move beyond simple data recording to uncover deep insights regarding high-performing states, crop yield efficiency, and seasonal variations.



Introduction

- Background and context of the project

Agriculture is the backbone of the Indian economy. However, the sector is complex, influenced by diverse climatic conditions, regional soil types, and varying seasonal patterns. This project is grounded in the "Indian Agricultural Crops Dataset," a massive repository of 73,827 records that captures the agricultural output of various Indian districts over several years. Understanding this data is essential because the sector contributes significantly to the GDP (over 18%), and optimizing it requires a shift from traditional observation to rigorous data analytics.

- Problem statement or goals of the project

The central problem this project addresses is the lack of accessible, visual insights into crop performance across the vast geography of India. Raw data alone is insufficient for identifying trends. The goal is to uncover key productivity insights, identify which states and crops are performing best, and determine where efficiency (yield per hectare) can be improved. Specifically, the project aims to identify the "High-Performing States" and "Top Performing Crops" to benchmark success.

- Scope and limitations of the project

The scope of this analysis is bounded by the attributes present in the dataset:

1. **Geographical Scope:** Analysis covers various States and Districts across India.
2. **Temporal Scope:** The data spans multiple "Crop Years," allowing for time-series analysis.
3. **Agricultural Scope:** It covers different seasons (Kharif, Rabi, Summer, Whole Year) and a variety of crops.
4. **Limitations:** The analysis is based on historical data provided in the dataset and relies on the accuracy of the reported "Area" and "Production" figures.

- Innovation component in the project

The innovative aspect of this project lies in the application of modern Business Intelligence (BI) tools to traditional agricultural data. By developing **interactive dashboards** in Power BI, the project moves away from static reports. This allows users to filter data dynamically—by season, state, or crop—to visualize correlations between "Area" and "Production" in real-time, facilitating a deeper level of exploratory data analysis (EDA).



Project Objectives

The project was executed with the following clearly defined objectives:

1. **Trend Analysis:** To systematically analyze crop production trends broken down by region (State/District), specific Crop types, and agricultural Seasons. This helps in understanding cyclical patterns in agriculture.
2. **Performance Identification:** To isolate and identify the "Top Performing Crops" based on production volume and the "Highest Producing States" to understand regional dominance in specific agricultural sectors.
3. **Correlation Visualization:** To visualize the relationship between "Total Cultivated Area" and "Total Production" using Power BI. This helps in assessing whether increased land usage actually leads to proportional increases in production.
4. **Dashboard Development:** To design and deploy interactive dashboards that allow stakeholders to drill down into the data, fostering an environment for data-driven decision-making.
5. **Policy Support:** To generate insights that are valuable for policymakers and researchers, enabling them to draft better agricultural policies based on historical performance and yield efficiency.

Expected Outcomes and Deliverables: The primary outcome of this project is the development of a fully functional, interactive Power BI dashboard that transforms over 73,000 raw agricultural records into dynamic visual narratives. This technical deliverable enables stakeholders to filter data by state, season, and crop year to uncover critical production trends and efficiency metrics in real-time. Beyond the dashboard, the project delivers a comprehensively cleaned and structured dataset suitable for future analytical research, alongside a detailed insight report that identifies high-performing states and top crops. Collectively, these outputs establish a robust decision-support framework, empowering policymakers and researchers to shift from static data observation to actionable, data-driven agricultural planning.



Methodology and Results

Methods/Technology used: The project followed a structured four-step roadmap to ensure data accuracy and analytical depth:

1. **Data Understanding & Cleaning:** The initial phase involved importing the 73,827 records and checking for inconsistencies, missing values, or data type errors to ensure the "RowID" and other attributes were clean.
2. **Exploratory Data Analysis (EDA):** Statistical analysis was performed to understand the distribution of data, such as the range of "Area" (hectares) and "Production" values.
3. **Dashboard Development:** Microsoft Power BI was used to model the data, create measures (DAX), and design the visual interface.
4. **Insights & Presentation:** The final phase involved interpreting the visuals to draw conclusions about agricultural trends.

Tools/Software used:

- **Microsoft Power BI:** The primary tool for data visualization and dashboard creation.
- **Data Source:** CSV/Excel Flat file containing the "Indian Agricultural Crops Dataset".

Data collection approach & Data Dictionary: The data is structured with specific attributes that allow for multi-dimensional analysis. The Data Dictionary used is as follows:

- **RowID (Integer):** A unique identifier for each record to ensure data integrity.
- **State_Name (String):** Enables state-wise performance analysis (e.g., "Bihar", "Assam", "Gujarat").
- **District_Name (String):** Allows for granular, district-level performance comparisons.
- **Crop_Year (Integer):** Crucial for tracking production trends over time (e.g., 2005, 2013, 2020).
- **Season (String):** Supports seasonal analysis (e.g., "Kharif", "Rabi", "Summer", "Whole Year").
- **Crop (String):** Identifies the specific type of crop cultivated (e.g., "Wheat", "Onion", "Maize").
- **Area (Float):** Measures the scale of cultivation in hectares.
- **Production (Float):** Measures the total output of the crop.



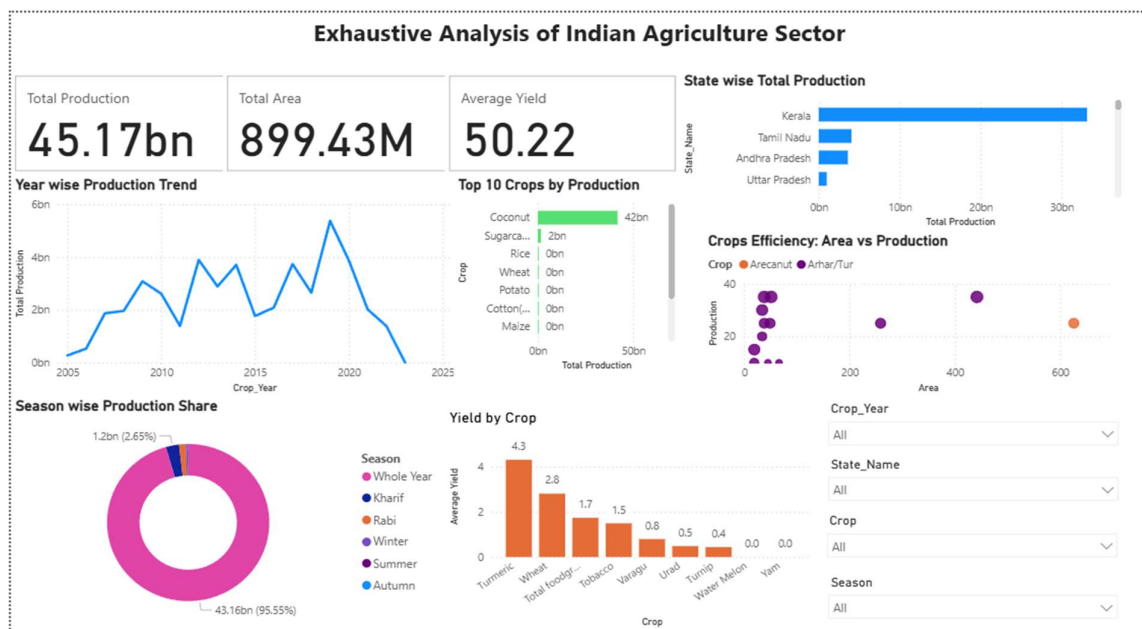
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Key Performance Indicators (KPIs) Modeled: To measure success, the following KPIs were calculated and visualized:

- **Total Production:** The sum of crop yields across all selected regions and years.
- **Total Cultivated Area:** The sum of agricultural land used, serving as the denominator for efficiency metrics.
- **Average Yield:** Calculated as the ratio of Production to Area. This is a critical efficiency measure indicating how productive the land is.
- **Top Performing Crop:** A ranking metric to identify which crop contributes the most to the national basket.
- **Highest Producing State:** A geographical metric to identify leading states in the sector.

Project Architecture: The architecture involved loading the raw dataset into Power BI Desktop, performing transformations in Power Query Editor (cleaning), creating a Star Schema (if applicable) or flat table Model, creating Calculated Measures using DAX for the KPIs listed above, and finally creating the Report View with charts and slicers.

Final project working screenshots along with supporting explanation:



Project GitHub Link:

<https://github.com/ankammabellamkonda/Ankamma-G19-SQL-POWERBI>

<https://github.com/sure-trust/BELLAMKONDA-ANKAMMA-g19-sql-and-powerbi>



Learning and Reflection

- Document every team member's new learnings (in terms of technology, management, etc)

During this internship, I expanded my technical skillset significantly across Data Analytics and Database Management:

1. **Data Visualization & BI Tools:** I learned to work with multiple industry-standard tools including **Power BI, Google Looker Studio, and Tableau**. Beyond just creating charts, I learned the art of **Data Storytelling**—selecting the right visuals to convey a narrative and creating professional presentations that drive decision-making.
2. **Advanced SQL & Database Concepts:** I gained deep technical knowledge in SQL, mastering concepts such as **Window Functions**, Joins, and DML operations (Insert, Delete). I also learned to distinguish between different database types like **PostgreSQL and NoSQL**, and understood the internal execution of queries.
3. **Data Architecture:** I learned how real-world companies architect their systems to handle **large-scale user databases**, ensuring data integrity and performance in a production environment.

Document every team member's experience (overall experience):

- **My Experience:** "My overall experience in this internship was immersive and practical. Learning directly from **instructors from the IT industry** provided me with real-world perspectives that go beyond textbooks. The learning environment was dynamic and enjoyable; we didn't just listen to lectures but engaged in **debates and hands-on projects** to implement our learnings immediately. This combination of technical rigor (in SQL and Power BI) and interactive learning has given me the confidence to handle real-world data challenges."



Conclusion and Future Scope

- Recap objectives and achievements

This project successfully achieved its objective of performing an exhaustive analysis of the Indian Agriculture sector. By processing 73,827 records, we developed a system that tracks production trends, identifies high-yield states, and monitors the efficiency of land use.

The Power BI dashboards created serve as a proof-of-concept for how data analytics can modernize traditional sectors. We successfully visualized the correlation between Cultivated Area and Production, providing a clear picture of agricultural efficiency.

- Future scope of this project

1. **Predictive Analytics:** The current project analyzes historical data. Future scope includes integrating Machine Learning (ML) algorithms to *predict* future crop yields based on historical trends.
2. **Weather Data Integration:** Agriculture is heavily dependent on weather. Integrating rainfall and temperature data would allow for a more robust analysis of climate impact on production.
3. **Market Price Analysis:** Combining production data with market price data (MSP) could help farmers and analysts understand the economic profitability of specific crops, not just the physical output.